Today's agenda

- No. of factors
- Prime numbers
- Sum of N natural no-s
- floor & ceil
- Sqrt
- .....

AlgoPrep

Q) Count no. of factors

↳ Given a number N, Print the count of factors.

N = 24 → {1 2 3 4 6 8 12 24} → 8

N = 36 → {1 2 3 4 6 9 12 18 36} → 9

```
P S V main () {
    Scanner scn = new Scanner (System.in);
    int n = scn.nextInt();
    int Count = 0;
    for (int i = 1; i <= N; i++) {
        if (N % i == 0) {
            Count ++;
        }
    }
}
```

iteration Count

N

By fact

1 sec = 10^8 iterations

No. of iterations = N

$N = 10^9$

10^9 iterations

10^8 iterations = 1 sec

1 iteration : $\frac{1}{10^8}$ sec

$1 * 10^9 = \frac{1}{\cancel{10^8}} * \cancel{10^9}^{10}$ sec

= 10 sec

$N = 10^{18}$ → Seconds?

↳ No. of iteration : 10^{18}

10^8 iterations = 1 sec

1 iteration : $\frac{1}{10^8}$ sec

10^{18} iteration : $\frac{1}{\cancel{10^8}} \times \cancel{10^{18}} = 10^{10}$ sec

10^{10} sec = 317 years → 10 sec

you → child → grandchild → 3rd → 4th → 5th gen

AlgoPrep

//optimize

$i * j = N \Rightarrow j = N/i$

$\Downarrow$

i & N/i are the factors

N = 24

i		j = N/i	
1	<	24	+2
2	<	12	+2
3	<	8	+2
4	<	6	+2
6	>	4	
8	>	3	
12	>	2	
24	>	1	

$i <= N/i$

$i^2 <= N$

$i <= \sqrt{N}$

N = 36

i		N/i	
1	<	36	+2
2	<	18	+2
3	<	12	+2
4	<	9	+2
6	==	6	+1
9		4	
12		3	
18		2	
36		1	

AlgoPrep

//Pseudo code

```
int CountFactors (int N){
    int Count=0;
    for (int i=1; i*i<=N; i++){
        if (N% i==0){
            if (i==N/i) { Count = Count +1; }
            else { Count = Count +2; }
        }
    }
    return Count;
}
```

No. of iteration

$\sqrt{N}$ iteration

N = 24

```
int Count=0;
for (int i=1; i*i<N; i++){
    if (N% i==0){
        Count = Count +2;
    }
}
```

N=24

Count = 0

i	i*i<N	N%i	Count (N/i)
1	+	+	2
2	+	+	4
3	+	+	6
4	+	+	8
5	b ↳ exit		

AlgoPrep

$N = 36$

```
int Count = 0;
for (int i=1; i*i<=N; i++){
    if (N% i == 0){
        if (i == N/i) { Count = Count + 1; }
        else { Count = Count + 2; }
    }
}
```

Count = 0

i	$i*i<=N$	$N\%i$	$i == N/i$	Count (N/i)
1	+	+	f	2 → 1 → 36
2	+	+	f	4 → 2 → 18
3	+	+	f	6 → 3 → 12
4	+	+	f	8 → 4 → 9
5	+	f		
6	+	+	+	9 → 6
7	f ↳ exit			

$\sqrt{N}$ iterations

↓

$N = 10^{18}$

⇓

No. of iterations $= \sqrt{10^{18}} = 10^9$

10^8 iterations = 1 sec

1 iteration $= \frac{1}{10^8}$ sec

10^9 iterations $= \frac{1}{10^8} \times 10^9$

$=$ 10 secs

Q) Prime numbers

↳ Given a number N, check if the number is a Prime no.

Prime numbers ⇒ Count of factors == 2

→ true ⇒ N is Prime
↳ false ⇒ N is non-Prime

No. of iteration
$\sqrt{N}$ iteration

```
boolean isPrime (int N){

    int Count = 0;
    for (int i=1; i*i<=N; i++){
        if (N% i == 0){
            if (i==N/i) { Count = Count +1; }
            else { Count = Count +2; }
        }
    }

    if (Count == 2){ return true; }
    else { return false; }
}
```

AlgoPrep

Quiz 1: Sum of all the numbers from 1 to 10.

↳ $\frac{10 * 11}{2} = \frac{110}{2} = 55$

Quiz 2: Sum of all numbers from 1 to 1000.

↳ $\frac{\overset{500}{\cancel{1000}} * 1001}{\cancel{2}} = 500500$

Q) Sum of first N natural numbers.

→ Gauss (4th class)

$$S = 1 + 2 + 3 + 4 + _ _ _ _ + 998 + 999 + 1000$$
$$S = 1000 + 999 + 998 + 997 _ _ _ _ _ + 3 + 2 + 1$$
$$2*S = 1001 + 1001 + 1001 + 1001 + _ _ _ _ + 1001 + 1001 + 1001$$
$$\Downarrow$$
$$1001 * 1000$$

$$2 * S = 1001 * 1000$$
$$S = \frac{1001 * 1000}{2} = 500500$$

// Sum of first N natural No-s

$$S = 1 + 2 + 3 _ _ _ _ _ _ + (N-2) + (N-1) + N$$
$$S = N + (N+1) + (N-2) _ _ _ _ _ + 3 + 2 + 1$$
$$(N+1) + (N+1) + (N+1) \qquad + (N+1) + (N+1) + (N+1)$$

$$2*S = (N+1)*N \Rightarrow S = \frac{N*(N+1)}{2}$$

Quiz 3: Sum of 1st N whole numbers.

$\hookrightarrow 0 + 1 + 2 + _____ + (N-1)$

$\Downarrow$

$1 + 2 + 3 + ___ + N-1 = \frac{(N-1) * (N-1+1)}{}$

$= \frac{(N-1) * N}{2}$

Sum of 1st N whole numbers = Sum of first (N-1) natural numbers.

Break till 9:40pm

AlgoPrep

floor(num) → just smaller or equal integer

Ex: 7.4 → 7
8.9 → 8
100.01 → 100
90 → 90
20.99 → 20
3 → 3

Math.floor(num);

AlgoPrep

Ceil(num) → just greater or equal integer

Ex: 7.4 → 8
8.9 → 9
100.01 → 101
90 → 90
20.99 → 21
3 → 3

Math.ceil(num)

AlgoPrep

Q) Given N, return floor (sqrt(N))

ex: N = 60 → 7..... → 7

N = 31 → 5.x → 5

N = 29 → 5.y → 5

N = 16 → 4.0 → 4

floor ($\sqrt{60}$) = 5 → 5*5 = 25

6*6 = 36

7*7 = 49

8*8 = 64 ××

```
int sqrt (int N){
    int ans = 1;
    int i = 1;
    while (i*i<=N) {
        ans = i;
        i++;
    }

    return ans;
}
```

N = 60

i	i*i<=N	ans = i
1	t	1
2	t	2
3	t (4*4<=60)	3
4	t (5*5<=60)	4
5	t (6*6<=60)	5
6	t (7*7<=60)	6
7	t	7
8	f (exit)	

↳ no. of iterations → $\sqrt{N}$

↳ you didn't come this for only to come this far.